Human-Factor-Coupled Collision Avoidance and Search-and-Rescue in a Multi-Agent Maritime Safety Simulation: A Baltic–North Sea Case Study

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Abstract

Between 70–96 % of maritime casualties involve human error [1], and heavy weather makes coordination harder by degrading both perception and bridge-to-bridge radio. We present a tick-based agent-based model of Baltic and North Sea traffic. Vessels follow depth- and lane-constrained routes built from open bathymetry and Automatic Identification System (AIS) density data. Encounters are resolved with a Closest Point of Approach (CPA) check and a COLREGs-style give-way rule. The main modelling choice is that every collision warning succeeds only with probability equal to the product of a local weather factor and a crew-fatigue factor. Collision damage uses DTU SIMCOL [2]; route-network collision frequency is checked with IWRAP Mk II [3]; rescue follows IAMSAR practice [4] with MASSIM/Koopman random search [5, 6] and Ashrafi seasonal degradation [7]. We compare a classical baseline, a weather-aware baseline, and a fatigue-aware Proposed System across four scenarios (30 seeds each; 360 runs) with Mann–Whitney U tests, Holm–Bonferroni correction, and Cliff's δ. Pre-registered fatal-rate and survival thresholds (H1–H3) were not confirmed after correction. Preparedness P_{prep} rose with large confirmed effects in every scenario ($|\delta| \geq 0.90$). Mean Storm Corridor collisions fell by about 28 % versus Baseline A (δ=0.32, not significant after Holm).

Index Terms: maritime safety, multi-agent systems, agent-based modelling, collision avoidance, human factors, search and rescue

1. Introduction

Maritime transport moves most of world trade, but safety remains limited by human factors [1]. The European Maritime Safety Agency (EMSA) [8] reports weather as a frequent compounding factor in serious casualties, especially in congested coastal waters, and hull and machinery claims remain large [9]. The International Regulations for Preventing Collisions at Sea (COLREGs) [10] assume that crews see the encounter and can agree who gives way. That assumption breaks down under two effects that many maritime ABMs treat only loosely [11, 12, 13]. Weather and radio: sea clutter and storms reduce the chance that a marine VHF warning is heard and acted on, yet many models assume that once risk is detected the manoeuvre always happens. Crew fatigue: fatigue builds over a watch and follows a circadian pattern [14], but prior work often keeps it in a side model without a direct link to whether a warning is executed. This paper couples those effects in one product: warning success depends on both local weather and crew fatigue (Fig. 1). Sections 3–6 define the Baltic–North Sea ABM, fatigue-aware CPA avoidance, SIMCOL damage [2], and SAR. Section 7 places network frequency on an IWRAP scale [3]; Sections 8–10 report the factorial comparison.

2. Objectives and Hypotheses

We test whether linking fatigue management to warning reliability reduces fatal outcomes more than classical, weather-unaware practice. We implement three configurations (storm-zone radio loss is shared by all methods—environmental, not a method privilege). Baseline A (Classical) uses COLREGs give-way with no weather-aware slowdown and unmanaged fatigue. Baseline B (Shore Rest Alerts) adds weather-aware speed reduction and slower fatigue build-up. The Proposed System keeps Baseline B's slowdowns and adds smart watch scheduling (lowest fatigue build-rate). Hypotheses: H1—Proposed reduces the fatal-event rate (Table 3, KPI 1) by $\geq$20 % versus Baseline A ($p < 0.05$, Mann–Whitney U [15], Cliff's $|\delta| \geq 0.2$ [16]); H2—Proposed improves post-incident survival (KPI 4) by $\geq$15 % versus Baseline A; H3—under Storm Corridor, Proposed beats Baseline B on the fatal-event rate ($p < 0.05$).

3. Simulation Model

3.1. Domain and Time

The domain is the Baltic and North Sea: latitude $\phi \in [50.5°, 66.0°]$ N and longitude $\lambda \in [-5.0°, 31.0°]$ E. An equirectangular map with a mid-latitude correction sets one field unit to one nautical mile at $\phi_0 = 58.25°$:

$$x = (\lambda - \lambda_{\min})\, k_{\text{lon}}, \qquad y = (\phi - \phi_{\min})\, k_{\text{lat}},$$

with $k_{\text{lat}} = 60\,\text{nm}/°$ and $k_{\text{lon}} = 60 \cos \phi_0$. East–west distortion at the box edges is about 6–10 % and is left uncorrected; distances inside the engine are Euclidean nautical miles. Coastline polygons from Natural Earth sit in an R-tree for land and grounding tests. Bathymetry is used only offline for route planning (Section 3.3). Time steps are $\Delta t = 15$ min (0.25 h); the default run length is $T = 2880$ ticks (30 days). A wall-clock variable starts at 06:00 UTC and drives the circadian fatigue term (Section 4). Each run uses one seeded `SmallRng`, so the same seed yields identical KPI logs.

3.2. Vessels and Voyage Logic

Agents are ships; after an incident, rescue assets appear (Section 6). Ship i has position $\mathbf{p}_i$, waypoint route R_i, travel direction $d_i \in \{+1, -1\}$, heading ψ_i, cruise speed v_i, state $\sigma_i \in \{\text{ACTIVE}, \text{DOCKED}\}$, crew $n_i = 10$, and fatigue $F_i \in [0, 1]$. Routes come from the offline network of Section 3.3 (`data/ais_paths.json`); ports are polygons in `data/ports.json`. Entering a port docks the ship. Cruise

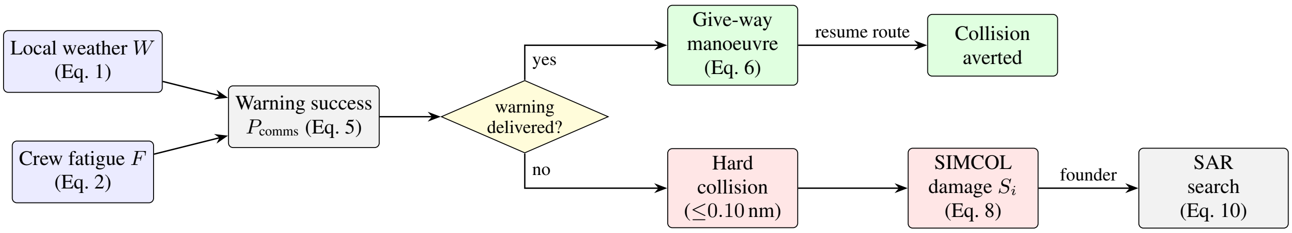

Figure 1: *Causal pipeline. Weather W and fatigue F set P_{comms}. A delivered warning yields a give-way manoeuvre; a missed warning leaves the encounter to geometry. Hard collisions feed SIMCOL; foundering enters SAR.*

speed is drawn at spawn ($u \sim \mathcal{U}(0,1)$):

$$v_i = \begin{cases} 15 + 8u & \text{passenger / ferry / ro-ro / ro-pax} \\ 10 + 4u & \text{tanker} \\ 10 + 6u & \text{cargo / container / bulk} \\ 10 + 10u & \text{otherwise.} \end{cases}$$

Waypoint pursuit: the active target is an avoidance waypoint if one is queued, otherwise the current route waypoint. A waypoint is reached within 0.5 nm. The step length is $s = \min(v_i\,\Delta t,\ \rho)$. Moves use up to five geometrically halved lengths so thin land is not tunnelled; a post-move check reverts grounding to the last valid position. At a port or route end the ship dwells $D \sim \mathcal{U}\{8, 48\}$ ticks, then reverses. Fleet size N is held fixed: when a ship is removed, a new one is spawned on a random synthesised track. Same-destination following: co-directional ships with endpoints within 5 nm share a destination; a trailer within 3 nm of a leader matches the leader's speed, and within 0.5 nm it holds station. This avoids side-by-side overlap that the give-way rule does not separate.

3.3. Route Generation

Traffic geometry is produced offline so experiments stay reproducible. Depth comes from the EMODnet DTM [17]. Lane attractiveness comes from EMODnet vessel-density maps [18, 12]. Both are fetched once via the OGC Web Coverage Service (WCS) and stored as grids. Ports are a gazetteer of 42 harbours snapped to navigable water. Each class c has draft T_c and under-keel clearance (UKC$_c$) [19]. A cell of depth h is navigable iff $h \geq T_c + \text{UKC}_c + m_s$ with buffer $m_s = 2$ m (Table 1). Deep-draft very large crude carriers (VLCCs) are excluded (Danish Straits).

Table 1: *Minimum transit depth by vessel class (draft + under-keel clearance).*

Class	Draft T_c (m)	UKC$_c$ (m)	Required h (m)
Passenger	6.5	1.5	8
Cargo	11.0	2.0	13
Tanker	14.0	3.0	17

Between sampled port pairs, a weighted A* planner [20] runs on a 1 nm grid. Cell cost uses

$$\mu = 1 + w_h\, s_h + w_d\, s_d + w_\ell\,(1 - a_\ell),$$

penalising shallow margins (s_h), proximity to shore/shoal (s_d), and off-lane cells (a_ℓ). Default weights are $(w_h, w_d, w_\ell) = (3, 4, 2.5)$. Paths are simplified under a depth constraint and checked against raw bathymetry every 0.15 nm. Seeded cost jitter keeps outputs deterministic per seed. The default network has 80 routes (35 cargo, 25 passenger, 20 tanker).

3.4. Weather Field

A 50 × 50 grid holds sea state s, visibility v, and wind u in $[0, 1]$. They fuse into hazard $W_{ij} \in [0, 1]$:

$$W = \text{clip}\left(\frac{w_s\, s + w_v\,(1 - v) + w_u\, u}{w_s + w_v + w_u},\ 0,\ 1\right), \tag{1}$$

with $(w_s, w_v, w_u) = (1.0, 0.5, 0.8)$. Each ship samples the nearest cell. The field advances each tick through a calm/storm regime chain, Lagrangian storm cells, AR(1) background fields, overlays, coupling, and smoothing (parameters in Appendix A). Independently, Storm Corridor places a moving storm of radius r_{storm} that drifts at 0.30 nm/tick along a fixed track (English Channel → North Sea → Skagerrak → Kattegat → Baltic → Gdańsk). Inside the zone, link reliability and non-A speeds are degraded (Sections 4 and 9.2; Table 2).

4. Crew Fatigue and Collision Avoidance

4.1. Fatigue Dynamics

Fatigue $F_i \in [0, 1]$ updates every tick. The circadian drive peaks near 03:00:

$$c(t_{\text{utc}}) = \tfrac{1}{2}\Big(1 - \cos\big(\tfrac{2\pi(t_{\text{utc}} - 3)}{24}\big)\Big) \in [0, 1].$$

While ACTIVE, with local hazard W,

$$\Delta F = \big(\beta_{\text{base}} + \beta_W\, W + \beta_c\, c(t_{\text{utc}})\big)\, m(\text{method}), \tag{2}$$

with $(\beta_{\text{base}}, \beta_W, \beta_c) = (0.0015,\ 0.0040,\ 0.0008)$ and

$$m(\text{method}) = \begin{cases} 1.00 & \text{Baseline A} \\ 0.82 & \text{Baseline B} \\ 0.65 & \text{Proposed,} \end{cases} \tag{3}$$

then $F_i \leftarrow \min(1,\ F_i + \Delta F)$. While DOCKED, recovery is 0.025 per tick. Above threshold $F^* = 0.40$, fatigue reduces warning reliability with maximum penalty $\kappa = 0.65$:

$$g(F_i) = \begin{cases} 1, & F_i \leq F^* \\ 1 - \kappa\, \dfrac{F_i - F^*}{1 - F^*}, & F_i > F^* \end{cases} \in [0.35,\ 1]. \tag{4}$$

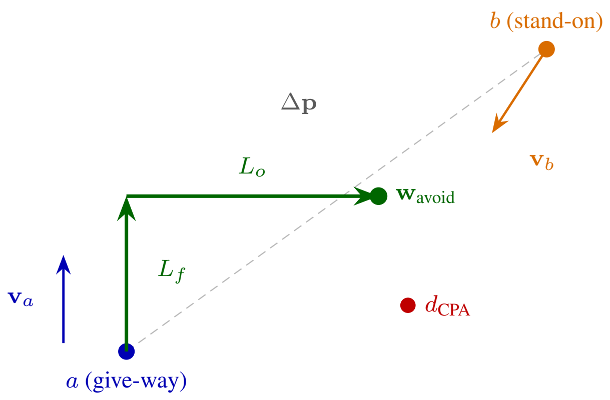

Figure 2: *Give-way geometry. Ship a steers to a temporary waypoint $\mathbf{w}_{avoid}$ placed L_f ahead and L_o to starboard (Eq. 6).*

4.2. CPA Detection and Give-Way

Each ACTIVE pair is screened with linear CPA. With relative position $\Delta\mathbf{p}$ and relative velocity $\Delta\mathbf{v}$,

$$t^* = -\frac{\Delta\mathbf{p}\cdot\Delta\mathbf{v}}{\|\Delta\mathbf{v}\|^2}, \qquad d_{\text{CPA}} = \big\|\Delta\mathbf{p} + t^*\,\Delta\mathbf{v}\big\|,$$

where t^* is in ticks. A manoeuvre is considered only if $\|\Delta\mathbf{p}\| \le 20\,\text{nm}$, $d_{\text{CPA}} \le 5\,\text{nm}$, $0 < t^* \le 12$ ticks, and the give-way ship is not in cooldown. The lower-id ship gives way (turn to starboard); the other stands on. This is a reproducible surrogate for COLREGs stand-on/give-way assignment [10, 11]. The manoeuvre runs only if the warning is delivered:

$$P_{\text{comms}} = \underbrace{\min\big(q(\mathbf{p}_a),\, q(\mathbf{p}_b)\big)}_{\text{weather / storm}} \cdot \underbrace{\min\big(g(F_a),\, g(F_b)\big)}_{\text{fatigue (Eq. 4)}}, \qquad (5)$$

where $q(\mathbf{p})$ is nominal link reliability, reduced to the storm-comms rate when $\mathbf{p}$ is in the storm zone (0.08 in Storm Corridor). A uniform draw below P_{comms} executes the manoeuvre; otherwise geometry alone decides the encounter. A give-way ship receives one temporary waypoint $L_f = 3\,\text{nm}$ ahead and $L_o = 5\,\text{nm}$ to starboard of heading ψ (Fig. 2):

$$\mathbf{w}_{\text{avoid}} = \mathbf{p} + L_f\,(\sin\psi,\ \cos\psi) + L_o\,(\cos\psi,\ -\sin\psi). \quad (6)$$

A short message log records the warning exchange; a 3-tick cooldown prevents thrashing.

4.3. Optional Force-Field Track

As an optional, default-off switch, the discrete waypoint of Eq. 6 can be replaced by a continuous repulsion field in the style of Xiao et al. [21]:

$$\mathbf{F}_{\text{rep}} = \sum_{k:\, d_k < d_o} \frac{k_{\text{rep}}}{d_k^p}\,\hat{\mathbf{n}}_k,$$

with exponent $p = 2$ by default. Source effect distances are micro-scale relative to a 15-minute basin tick, so magnitudes are rescaled while keeping the reported head-on to overtaking ratio. All definitive experiments keep this track off.

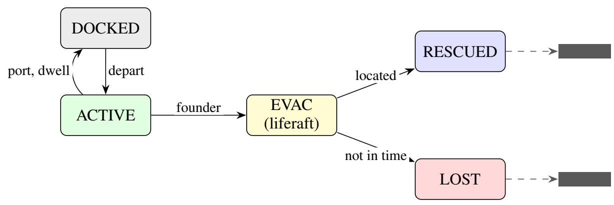

Figure 3: *Vessel state machine. Foundering collision → EVAC; SAR yields RESCUED (Eq. 10) or LOST. Terminal ships respawn to keep fleet size.*

5. Collision Consequence (SIMCOL)

A **hard collision** is logged when two ACTIVE ships close within 0.10 nm (≈185 m), once per encounter (rising edge). Contacts within 1 nm of a port are ignored as harbour manoeuvring under vessel traffic services (VTS). Damage uses DTU SIMCOL (ship collision damage / survivability) [2]. Absorbed energy follows Pedersen and Zhang [22, 2] on the normal closing component:

$$E_{\text{abs}} = \tfrac{1}{2}\,\frac{M_a' M_b'}{M_a' + M_b'}\,V_n^2, \qquad V_n = \big|(\mathbf{v}_b - \mathbf{v}_a)\cdot\hat{\mathbf{n}}\big|, \quad (7)$$

with added-mass factors $m_{\text{sway}} = 0.85$ and $m_{\text{surge}} = 0.05$ [2]. Minorsky's correlation [23] links energy to damaged volume. Relative penetration t/B drives a survival factor shaped like SOLAS Chapter II-1 probabilistic damage stability [24]:

$$S_i = \begin{cases} 1, & t/B \le r_0 \\ \left(\dfrac{r_1 - t/B}{r_1 - r_0}\right)^{0.25}, & r_0 < t/B < r_1 \\ 0, & t/B \ge r_1, \end{cases} \qquad (8)$$

with $r_0 = 0.10$, $r_1 = 0.30$. This is an accepted surrogate: the engine does not carry per-ship residual-stability curves. Expected overboard count is

$$\mathbb{E}[\text{overboard}] = n_b\,\big(1 - S_i\big), \qquad (9)$$

drawn as independent Bernoulli trials. If $S_i < S^\dagger = 0.50$, the struck ship founders and the whole crew enters the liferaft (EVAC, Section 6). If it stays afloat, only the overboard draw enters the man-overboard chain (Section 6.3). Encounter type is diagnostic, from the folded heading angle θ:

$$\text{type} = \begin{cases} \text{side-to-side (overtaking)}, & \theta < 45^\circ \\ \text{front-to-side (crossing)}, & 45^\circ \le \theta < 135^\circ \\ \text{head-on}, & \theta \ge 135^\circ. \end{cases}$$

Near-parallel hits release little energy in Eq. 7; head-on and crossing hits drive deeper penetration.

6. Search and Rescue

Foundering starts the liferaft chain under International Aeronautical and Maritime Search and Rescue (IAMSAR) Manual practice [4]. A rescue asset leaves the nearest port after a mobilisation delay of 2 ticks by default: helicopter (80 kn) beyond 40 nm offshore, otherwise patrol (25 kn). Persons overboard from afloat ships use a separate chain (Section 6.3). Figure 3 shows the vessel state machine. Grounding only reverts position; it does not force evacuation.

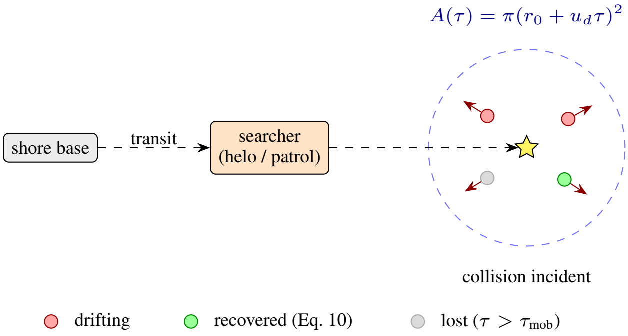

Figure 4: *Man-overboard search. Casualties (Eq. 9) drift separately; one searcher applies Eq. 10 per person.*

6.1. Drift and Detection (MASSIM)

Detection follows the Koopman random-search model used in the MASSIM maritime-search simulation [5, 6]. With sweep width w, search speed v_s, and area A,

$$\mathrm{CDP} = 1 - \exp\left(-\frac{n\,w\,v_s\,\tau}{A}\right) = 1 - e^{-C}. \qquad (10)$$

Search area grows as $A(\tau) = \pi\,(r_0 + u_d\,\tau)^2$ with initial radius r_0 and drift u_d. Each on-scene tick draws Bernoulli detection at coverage $1 - e^{-\mathrm{d}C}$.

6.2. Seasonal Degradation (Ashrafi)

Asset performance follows Ashrafi et al. [7]. Wind-chill uses

$$\mathrm{wci} = 13.12 + 0.6215\,T - 11.37\,U^{0.16} + 0.3965\,T\,U^{0.16}.$$

Months map to severity groups A–D (May–August mildest; Dec–Feb harshest). Relative transit/search speeds are approximately $1.00/0.95/0.77/0.60$ (helicopter) and $1.00/0.80/0.40/0.21$ (patrol); boarding times lengthen with severity [7]. Default simulation month is July (group A).

6.3. Man-Overboard Search

Overboard casualties (Eq. 9) form one incident of persons-in-water, each with its own drift bearing (Fig. 4), following Karatas et al. [5]. One searcher homes on the cluster centroid and applies Eq. 10 independently per person. Anyone not recovered within $\tau_{\mathrm{mob}} = 12$ ticks (≈3 h) is lost. Liferaft endurance is 96 ticks (24 h).

7. External Collision-Frequency Check (IWRAP)

Internal KPIs compare methods. Network collision frequency is checked with IALA's IWRAP Mk II [3, 25]:

$$N_c = N_a\,P_c.$$

Head-on, overtaking, and crossing candidate counts follow the usual IWRAP forms (flows Q, beams B, speeds V, Pedersen diameter for crossings [3, 26]). Causation probabilities are the IWRAP defaults: $P_c^{\text{head-on}} = 0.50 \times 10^{-4}$, $P_c^{\text{overtaking}} = 1.10 \times 10^{-4}$, $P_c^{\text{crossing}} = 1.30 \times 10^{-4}$. Each batch row stamps N_c from the static synthesised routes, the scenario fleet size, and a method speed scale (weather-aware methods slow traffic). We do not export per-run AIS tracks. For the present network, N_c is on the order of 0.01 collisions/yr, inside the usual $<$1/yr/fairway acceptance band.

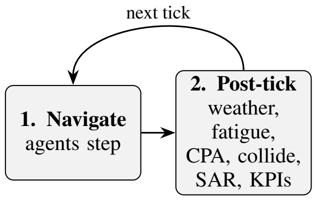

Figure 5: *Macro-tick order. Agents navigate, then the post-tick pipeline advances weather, fatigue, CPA/comms, collisions, SAR/MOB, fleet top-up, and KPIs.*

8. Experimental Design

8.1. Methods and Scenarios

The three methods of Section 2 differ only by fatigue build-rate m (Eq. 3), weather/storm speed awareness (Section 9.2), and preparedness weight α_M below. SIMCOL parameters are shared. Table 2 lists the four scenarios. Each condition uses 30 seeds and 2880 ticks (360 runs).

8.2. Key Performance Indicators

Table 3 defines the six hypothesis KPIs. Ship-hours add 0.25 h per ACTIVE ship per tick. A KPI *fatality* is an unrecovered person-in-water within τ_{mob} (Section 6.3); liferaft LOST is logged separately and does not enter KPI 1. Crew preparedness uses method awareness $\alpha_M \in \{2.0\,(\mathrm{A}),\ 1.3\,(\mathrm{B}),\ 1.0\,(\mathrm{Proposed})\}$:

$$P_{\mathrm{prep}} = \max(0,\ 1 - \alpha_M W) \cdot \max(0,\ 1 - 0.70\,F).$$

Diagnostic counters (collision-type mix, MOB tallies) are logged but not entered into the Holm battery.

8.3. Statistical Analysis

For each (KPI × scenario × method pair) we use a two-tailed Mann–Whitney U test [15] with Cliff's δ [16]. Multiplicity uses Holm's sequentially rejective Bonferroni procedure [27] over the full battery. A result is *significant* if $p_{\mathrm{corr}} < 0.05$ and *confirmed* if also $|\delta| \geq 0.2$. Pairs are (A vs Proposed), (B vs Proposed), and (A vs B). Seeds are matched across conditions.

9. Implementation

The engine is Rust on krABMaga [28]. An Axum HTTP API launches factorial batches (Rayon) and streams progress with Server-Sent Events (SSE). Each finished run writes a JSONL tick log (every ten ticks), a manifest, and a SQLite results row. A React workbench (deck.gl / MapLibre) replays completed runs.

9.1. Tick Ordering

Within a tick, vessel agents step first (krABMaga order 200), then a post-tick agent (order 1000) runs the shared physics pipeline in Fig. 5; navigation therefore uses weather and avoidance state from the previous tick's post-tick pass.

9.2. Speed Reduction

Baseline B and Proposed interpolate the just-moved position toward the last valid position by

$$f_W = \max\left(0.30,\ 1 - 0.45\,W\right).$$

Table 2: *Scenario summary (30 seeds, 2880 ticks/run).*

Scenario	Key conditions	Ships	Hyp.
Calm Passage	Calm weather; calibration reference (target band ≈0.5–1.5 collisions / 1 k ship-hrs under Baseline A).	20	Calib.
Storm Corridor	Moving storm; storm-comms rate 0.08; non-A speed factor 0.45.	25	H1, H3
Blind Shore	Fleet-wide comms 0.55; tighter avoidance berth ($L_o = 3\,\text{nm}$).	25	H1
Deep Water Rescue	Sparse fleet; SAR mobilisation 4 ticks; helo/patrol $55/16\,\text{kn}$.	15	H2

Table 3: *KPI definitions and hypothesis links.*

#	KPI	Definition	Hyp.
1	`fatal_per_1k_hrs`	Fatalities ÷ ship-hrs $\times 10^3$	H1, H3
2	`collision_per_1k_hrs`	Collisions ÷ ship-hrs $\times 10^3$	Calib.
3	`evac_activation_rate`	Evac activations ÷ ship-hrs $\times 10^3$	—
4	`survival_ratio`	$1-$ fatalities / crew exposed in collisions	H2
5	`avg_tta_hours`	Mean rescue time-to-arrival	H2
6	`mean_p_prep`	Time-averaged crew preparedness	mech.

Table 4: *Pre-registered hypothesis outcomes (definitive 30-seed batch).*

Hyp.	Contrast	Verdict	Cliff's δ
H1	Fatal rate, Proposed vs A	Not confirmed	$\lvert\delta\rvert \le 0.07$
H2	Survival ratio, Proposed vs A	Not confirmed	$\lvert\delta\rvert \le 0.07$
H3	Fatals under Storm, Proposed vs B	Not confirmed	$\delta \approx 0.00$

Baseline A ignores this factor. Inside the storm zone an extra factor applies: 1.0 for Baseline A, 0.45 in Storm Corridor for the other methods (default non-corridor storm factor 0.70 when a storm is enabled).

10. Results

10.1. Calibration and Storm Contrast

Confirmation criteria are those of Section 8.3. Figure 6 shows per-run KPI distributions; Figure 7 summarises effect sizes for the pre-registered contrasts. Under Baseline A, Calm Passage mean collisions were 0.60 per 1 000 ship-hours (inside the 0.5–1.5 band). Storm Corridor raised that rate to 1.48 (2.48× calm), restoring a clear weather contrast after the depth+lane route rebase.

10.2. Hypotheses H1–H3

Table 4 summarises the pre-registered contrasts. **H1** and **H2** are not confirmed: fatal rates and survival ratios sit near floor/ceiling in Calm Passage, Storm Corridor, and Blind Shore, so tests have little power. Deep Water Rescue has a higher fatal base rate; Proposed reduces mean fatals from 0.038 (Baseline A) to 0.023 (≈39 %), but survival is almost unchanged and neither contrast survives Holm ($p_{\text{corr}} = 1$, $\lvert\delta\rvert \le 0.02$). **H3** is non-significant (means 0.0022 vs 0.0024).

10.3. Mechanism and Collisions

Mean P_{prep} is higher for Proposed than for either baseline in every scenario, with confirmed effects ($\lvert\delta\rvert \in [0.90, 1.00]$); mean Storm Corridor collisions fall from 1.48 to 1.07 versus Baseline A (≈28 %, δ=0.32), but that contrast is not significant after Holm.

11. Discussion

The core claim is Eq. 5: a warning succeeds only when weather and both crews allow it. The batch confirms the preparedness lever and a directional storm-collision benefit, but not the pre-registered fatal/survival thresholds. Fatal events are rare outside Deep Water Rescue, so H1–H3 are under-powered even when point estimates move in the expected direction. The survival factor S_i (Eq. 8) is a SOLAS-shaped surrogate, not a full residual-stability calculation. Main limitations: synthesised routes trade some traffic realism for control; the lower-id give-way rule simplifies COLREGs; force-field avoidance is off by default; IWRAP stamps use static routes rather than per-run tracks. Calm Passage calibration under Baseline A sits near published AIS near-miss bands [12, 13]. Longer runs or collision-rate primary endpoints would raise power on fatal contrasts.

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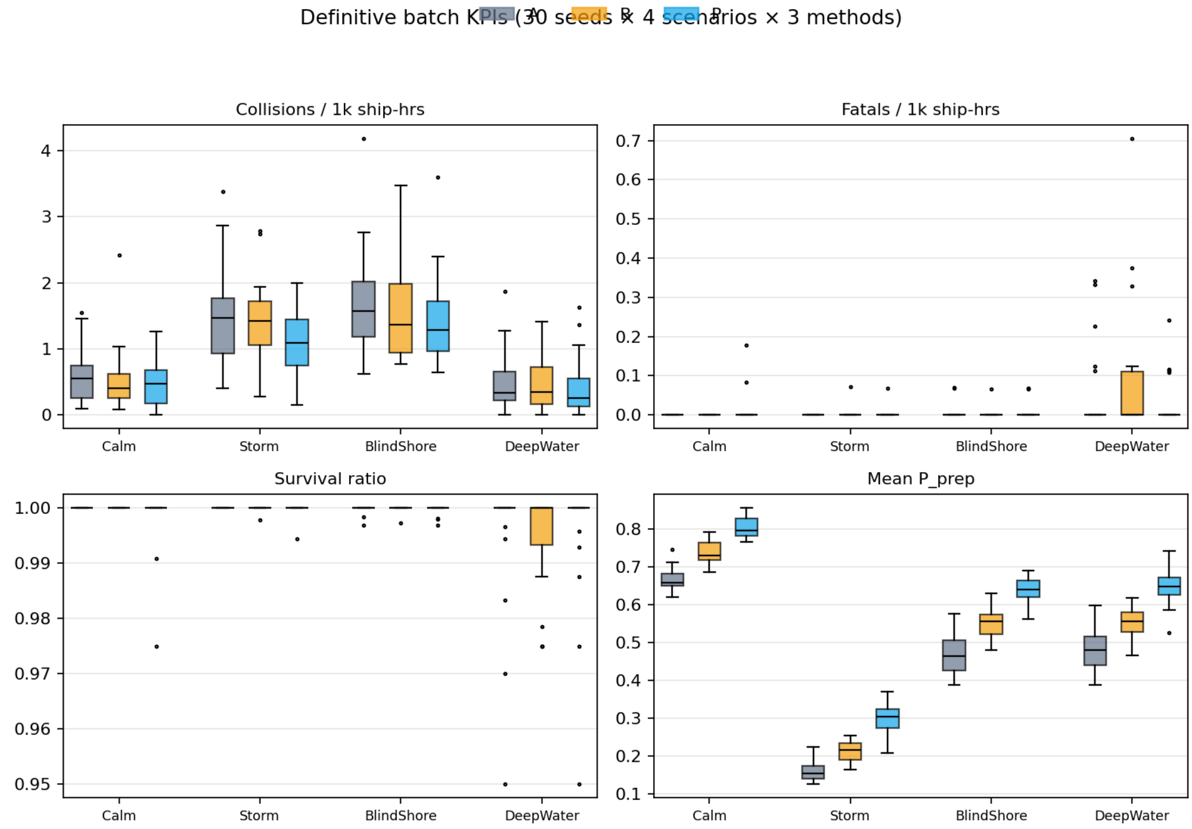

Figure 6: *Per-run KPI distributions for the definitive batch (30 seeds). A = Baseline A, B = Baseline B, P = Proposed.*

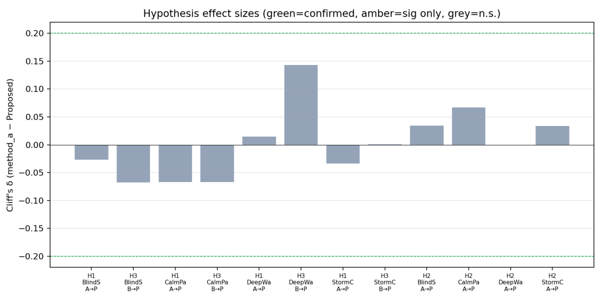

Figure 7: *Cliff's δ for H1–H3-relevant pairs. Green: confirmed; amber: significant only; grey: not significant after Holm.*

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A. Default Constants

Table 5 lists principal defaults (overridable in configuration).

Weather-field process (Mixed preset): regime flips $p_{cs} = 0.03$, $p_{sc} = 0.08$; Calm/Storm reversion targets $(0.30, 0.80, 0.25)$ / $(0.72, 0.30, 0.75)$; up to 8 storm cells; AR(1) coefficients $(\rho_s, \rho_v, \rho_u) = (0.95, 0.90, 0.93)$, $(\sigma_s, \sigma_v, \sigma_u) = (0.07, 0.06, 0.07)$; gradient cap 0.12. Calm and Stormy presets scale the flip rates.

Table 5: *Selected engine default constants.*

Constant	Default	Role
Δt	0.25 h	tick length (15 min)
T	2880	ticks per run (30 d)
weather grid	50×50	hazard field
(w_s, w_v, w_u)	$(1.0, 0.5, 0.8)$	hazard fusion weights
warn / CPA / TTA gate	20 / 5 nm / 12 tk	avoidance gate
L_f, L_o	3 / 5 nm	give-way waypoint offsets
cooldown	3 tk	re-manoeuvre lockout
r_{coll}	0.05 nm	hit at $2r = 0.10$ nm
port collision exclusion	1 nm	harbour water
follow vicinity / gap	3 / 0.5 nm	speed-match / keep-distance
same-destination	5 nm	endpoint match radius
$(m_{\text{sway}}, m_{\text{surge}})$	$0.85, 0.05$	SIMCOL added mass
Minorsky a, b	$47.2, 32.7$	energy–volume (MJ, m^3)
$\ell/L_{pp}, \tau$	0.12, 0.08 m	damage length / side steel
$r_0, r_1, S^{\dagger}$	$0.10, 0.30, 0.50$	S_i knee / loss / founder
$(\beta_{\text{base}}, \beta_W, \beta_c)$	$0.0015, 0.0040, 0.0008$	fatigue rates
recovery	0.025/tk	dock fatigue recovery
F^{*}, κ	0.40, 0.65	comms-degradation knee
storm-comms (corridor)	0.08	Storm Corridor link rate
storm-comms (generic default)	0.30	when storm enabled outside corridor presets
n crew	10	per vessel
dwell	8–48 tk	port stay
helicopter / patrol	80 / 25 kn	rescue asset speeds
Δt_{mob}	2 tk	rescue mobilisation delay
helo / patrol cutoff	40 nm	SAR asset-selection range
w, r_0, u_d	5 nm, 2 nm, 0.2 nm/tk	MASSIM sweep / datum / drift
survival window	96 tk	liferaft endurance (24 h)
τ_{mob}	12 tk	MOB cold-water window (3 h)
MOB scatter	0.5 nm	person-in-water spread
P_c (IWRAP)	0.5–1.3×10^{-4}	causation probability